

0013 — RevFT i1R / PB Retest Trade: Full Grid Study & Why It Fails — 2026-07-07

****Series:**** MC Setup Research Notes · Note 0013 ****Confidence:**** High ****as a negative result**** — two independent signal exports (6.4k and 12.7k signals), 5 years, tick-level event ordering, 72-cell exit grids, dose-response across three arming variants. Exploratory (no pre-registration), but the conclusion is the same in every cut.

TL;DR: We built the "rev-bar traders' breakout test": when a RevFT reversal runs to the rev-bar traders' +1R without an AI Brooks pullback (the **i1R** condition), we fade the pullback back into their entry price E (rev-bar extreme +/-1t), confirmed by a bar that touches E and closes inside a 1xABR(8) box. It fires ~24% of the time (1,148 trades on the 6k export; 2,852 on the 12k export) and is **flat net of costs in every exit configuration tested** — 6 stop modes x 12 target modes, both files. Stricter "cleaner run" arming makes it worse, monotonically. The one positive pocket (BO/IB with structural targets) shrinks on the larger file. Mechanism: **the cleaner the initial reversal, the more its retest of E marks the end of the move, not a defended pullback** — the rev-bar traders are paid and gone; there is nobody left to defend E. We also close a side door for good: the rev-bar stop-entry cannot be measured from confirmed-signal exports (survivorship: signal bars close beyond the break level 92.8% vs 44.3% base rate — the failed breaks a real stop order would have filled are missing from the file).

1. The setup (so this note stands alone)

Signals. RevFT ("MyReversals") 5M ES swing-reversal signals. Each csv row: signal bar (close-stamped), direction, and StopPrice = the **setup extreme** — the exact high/low of the swing the reversal formed off (verified: matches a bar extreme exactly in 479/479 sampled signals). Two exports: the standard RTH file (6,442 signals, 2021-06 to 2026-07) and the "All / Nymex Energy RTH" file (12,707 signals, different session template, only 3,642 overlap). Sneaky excluded throughout. The base RevFT trade at 1:1 is a firm loser (-0.09R; see note 0005) — this study is another rescue attempt, via a different entry on the same information.

Definitions (short side; longs mirrored).

- **Reversal bar** = the bar before the signal bar. **E** = reversal-bar low - 1t = the price where Brooks-style rev-bar traders went short on a stop.
- **1R range** = (setup extreme + 1t) down to E. **1R target** = E - range: the rev-bar traders' +1R.
- **i1R condition (arming)**: the 1R target is taken out — by the signal bar itself, or later — with **no AI Brooks pullback** on the way (no bar trading above the prior bar's high, tick-ordered inside bars).
- **PB signal**: after arming, the first bar that touches E and **closes inside a box of 1.0 x ABR(8)** centered on E (ABR(8) = mean H-L of the 8 bars before the touch bar). The setup extreme must never trade in between (rule 11). Entry: sell at the touch bar's close (fill = first tick of next bar, 1t slip, \$5 RT, EOD flat).

All event ordering (target takeout vs pullback vs extreme takeout vs E-touch) is resolved on **raw ticks**, not bars — this matters: a bar-level version misclassifies ~3% of arms (a bar can make its higher high after the target traded).

2. The questions

1. Does the PB retest trade make money under any reasonable stop/target scheme? 2. Does it replicate on the second, larger signal export? 3. Does stricter arming (a cleaner initial run) improve the retest trade? 4. Can the rev-bar traders' own trade (stop entry at E on the break) be evaluated from these files?

3. How we tested it

- **Exit grid per fired trade**: stops = S1 extreme+1t · S2 fill+/-{0.5,1,1.5}xABR(8) · S3 "AR" (start S1, tighten to entry-bar extreme+1t at its close) · S4 PB-bar extreme+1t. Targets = T1 the original csv trade's 1R level (fallback 1x stop) · T2 {0.5,1,1.5}xABR(8) · T3 {1..5}xAR (armed after entry bar closes) · T4 {1,2,3}x stop size. 72 combos, all tick-simulated, R reported against each row's initial stop distance.
- **Replication**: identical pipeline on the 12k export (11,017 signals mappable onto RTH bars).
- **Dose-response**: three arming variants — (a) the i1R rule above; (b) gate additionally on the csv trade winning its own farther 1R cleanly; (c) zero-MAE: price never re-crosses E at all before the target.
- **Side door**: stop-entry at E during the break, plus a causality audit of the export itself.

4. Results

4.1 The PB trade is flat — both files

	6k file	12k file
PB trades fired	1,148 (~24% of signals)	2,852 (~26%)
Baseline exit (S1 stop, T1 target)	+0.001R, 37% win, +\$2,648	+0.002R, 37% win, +\$17,315
Best cell of 72	+0.033R +/-0.087	+0.062R +/-0.059 (S2_1.5 x T4_3)
Cells meaningfully positive	none (best-of-72 selection)	none after selection discount

Grid structure is identical in both files: tight stops (S2_0.5, S4) lose everywhere — the retest zone is noisy; the structural target (T1) and large multiples of wide stops do best; AR-multiple targets (T3) are the worst family (-0.1 to -0.5R); longs = shorts.

Year-split of the 12k best cells: S2_1.5xT1: 2021 -0.00, 2022 +0.07, 2023 +0.00, 2024 +0.02, 2025 +0.01, 2026 +0.12. Nothing stable enough to trade; the positive mean leans on 2022/2026.

4.2 By setup type — the BO/IB pocket shrinks on the bigger file

Type	6k: n / S1xT1	12k: n / S1xT1	12k best cell
BO	148 / +0.113 +/-0.227	470 / +0.025 +/-0.121	+0.170
IB	245 / +0.086 +/-0.179	599 / +0.010 +/-0.115	+0.123
OB	128 / -0.037	371 / -0.085	-0.002
Trap	627 / -0.051	1,412 / +0.013	+0.056

The 6k file's one hopeful pocket (BO +0.11, best cell +0.28) **regresses toward zero at 3x the sample**. That is the signature of noise, not of a small edge.

4.3 Dose-response: the cleaner the run, the worse the retest

Arming (strictness increasing)	n	Baseline outcome
i1R (rules as specified)	1,148	~ = 0.00R
+ csv trade must win its own 1R cleanly	329	-0.09R, all 72 cells negative
Zero-MAE (never re-crosses E before target)	65	-0.2 to -0.4R

Monotone. This is the study's most informative pattern (discussed in section 5).

4.4 The side door, closed: rev-bar stop entry is unmeasurable from these files

A stop order at E during the break looks spectacular from the csv (+0.3 to +0.8R everywhere, tight stops "best") — and is entirely survivorship. The export's signal bars close beyond the break level **92.8%** of the time vs **44.3%** for all prior-bar-extreme breaks (n=90,366): a row only exists when the break worked into the bar close. The failed breaks — which a resting stop order fills all the same — are absent. Mean conditioned "head start" at signal-bar close: +0.39R, i.e. the entire apparent edge. Only 13 of 4,849 fills were causally placeable (order at signal-bar close). Same bias family as the ER10 bug (note 0002). A causal variant — LIMIT order at E placed at signal-bar close — fills 4,274 times and is **flat** (best cells ~+0.03R), consistent with the PB grid.

5. Why the i1R PB trade doesn't work as hoped

The trade's premise: rev-bar traders entered at E, they are +1R, so a pullback to E should be defended — by their re-entries, by breakeven stops sitting there, by late traders wanting the "second chance" entry. The data says the premise fails, and the dose-response says why:

- The defenders are gone by the time we need them.** A clean, pullback-free run to +1R is precisely the move where early shorts take profits (the 1R target is their exit) and trailing entries never got a pullback to join. When price finally returns to E, the cohort with an interest in defending it has largely monetized. The retest is not a test of committed inventory — it is price re-entering a vacuum.
- Selection cuts against us, monotonically.** If shallow-retest-then-continue were the dominant behavior, stricter cleanliness should help (purer momentum, stronger defense). It does the opposite at every step (0.00 -> -0.09 -> -0.3R). The cleaner the impulse, the more often the return to E is a genuine V-reversal that runs straight through the level. Brooks' own framing agrees: strong spikes get one pullback that holds well above the entry price; price actually coming all the way back to the breakout price after a clean 1R move is more often "failed breakout" behavior than "breakout test" behavior on this timeframe.
- The information is stale by construction.** Everything the setup knows (extreme, E, the clean run) is public an hour before our entry. The market has already paid out the reversal move once; our entry buys the second serving of an edge that note 0005 showed doesn't exist even for the first serving. Adding costs (~\$17/trade at median 8-pt risk) on top of a ~0 gross means the trade needs the retest to be defended more often than randomly — and it is defended slightly less often.
- The exits can't rescue an entry with no edge.** 72 stop/target geometries on two files move the needle from -0.02 to +0.06R — exactly the spread you get re-slicing noise. The only structure (wide stops > tight stops, structural targets > volatility targets) says the level is noisy, not that it is an edge.

6. Verdict & what survives

- **Retire the PB retest trade** as specified, all variants, both signal files. Do not re-tune exits; the entry has no edge to manage.
- **The framework survives:** tick-ordered arming/retest/box detection, the 72-cell exit grid, and the per-trade chart library are reusable on any signal set with a real edge (e.g. the MC signals).
- **The survivorship audit (4.4) is the durable lesson:** any "trade the setup bar's break" idea backtested from a confirmed-signal export will look brilliant and be fake. The honest route to the rev-bar entry question is a candidate-level export (setups stamped before confirmation) — the LizardTrader Auction Bars indicator documents exactly such candidate

series; exporter built (nt8/indicators/LTReversalCandidateExporter.cs), vendor series currently NULL on chart (pending settings/vendor resolution), or we re-implement the manual's mechanical candidate definitions directly on our bars.

Artifacts: docs/living/pb_grid_trades.parquet (+_12k, _i1r, _defA), limitE_grid_trades.parquet, revbar_grid_trades.parquet, i1r_flags.parquet, chart library docs/living/pb_trade_library/{longs,shorts}/index.html, verification charts docs/living/pb_verify/.